CURRICULUM VITAE

Name: Dr. SAIKAT BATABYAL
Date of birth: 05/24/1991
Nationality: Indian
Marital status: Married
Sex: Male
Address (present): Department of Biotechnology, Faculty of Life & Allied Health Sciences, Ramaiah University of Applied Science, Bangalore, India
Address (permanent): Bolpur, Kalikapur, West Bengal, India, 731204
Email: stbatabyal@gmail.com

DEGREE EARNED:

1. **Ph.D. (Applied Mathematics),** SRM Institute of Science and Technology, Chennai, India, **2022.**
2. **M.Sc. (Mathematics),** Visva Bharati Central University, India, **2015.**
3. **B.Sc. (Mathematics),** Visva Bharati Central University, India, **2013.**

HIGH SCHOOL DIPLOMA:

1. **12th Standard (Science),** West Bengal Council of Higher Secondary, India, **2009.**
2. **10th Standard (General Science),** West Bengal Board of Secondary Education, India, **2007.**

PRESENT DESIGNATION: **Assistant Professor**, Department of Biotechnology, Faculty of Life & Allied Health Sciences, Ramaiah University of Applied Science, Bangalore, India, 560054.

EXPERIENCE:

1. **Postdoctoral Scientist,** Texas Biomedical Research Institute, San Antonio, USA, **2023-2026.**
   - Key responsibility: Focusing on developing mathematical models of the within-host dynamics of pathogens, such as TB, and investigating the dynamics of the immune response to pathogens.

2. **Post-Doctoral Research Associate,** University of Tennessee, Knoxville, USA, **2023.**
   - Key responsibility: Mechanisms of Mtb dissemination in the lung, lymph node and spleen.

3. **Policy Researcher,** Indian Institute of Science (**IISc**), Bengaluru, India, **2021-2023**.
   - Key responsibility: Policy Research on Science Technology and Innovation.

4. **Senior Scientist,** University of Klagenfurt, Austria, **2017-2018.**
   - Key Responsibility: Sparsity-promoting variational regularization for linear inverse problems.

5. **Visiting Researcher,** Jacobs University of Bremen, Germany, **2015.**
   - Key Responsibility: Complex dynamic systems.

COMPUTING SKILLS:

R, Python, MatLab, Latex, Fortran, C Language, Microsoft Office, Excel.

RESEARCH INTEREST:

Immunology, Mathematical Modeling, Epidemiology, Computational Biology.

THESIS TITLE:

Finite Time Blow Up in a Realistic Food Chain Model and Control Population Explosion.

PROJECT DURING M. Sc. COURSE:

Project Title: Hyper Relative Order of Entire Functions with respect to Composite Entire Functions.

AWARDS:

1. **Oral Presentation Award**, Vaccine Development Center of San Antonio Conference, **USA, 2025**.
2. **2nd Place in Poster Presentation**, Texas Biomed Research Symposium, **USA, 2025**.
3. **IN-TRAC Travel Grant**, Texas Biomedical Research Institute, San Antonio, **USA, 2025**.
4. **Landahl Travel Award** by Society for Mathematical Biology (**SMB**), **2024**.
5. **Modeling Simulation Optimization** (MSO) Fellowship, **Austria, 2017**.

ACHIEVEMENTS:

1. **Adjunct Faculty**, Theoretical Immunology Laboratory, Texas Biomedical Research Institute, San Antonio, **USA, 2026**.
2. **Science Technology Innovation Policy** (STIP) Secretariat, Policy Coordination & Programme Management (PCPM) Division, Department of Science & Technology (DST), **Ministry of Science & Technology, Government of India, New Delhi, India, 2021-2023**.

3. **Member, National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS)**, Frontier and Futuristic Technologies (FFT) Division, **DST, Ministry of Science & Technology, Government of India, New Delhi, India, 2021-2023**.
4. **Member Secretary,** Uttar Pradesh Council of Science & Technology (**UPCST**)-**DST, Govt. of India, 2022**.

CONFERENCES/SYMPOSIA ATTENDED:

1. Vaccine Development Center of San Antonio Conference, 2025, United States.
2. Texas Biomed Research Symposium, 2025, San Antonio United States.
3. International Conference on Mathematical Biology, Quantitative Modeling and Computation in Life Sciences and Medicine, June 12-14, 2025, Pokhara, Nepal.
4. 2025 Texas Biomed Research Symposium, Texas Biomedical Research Institute, April 4, 2025.
5. 2025 Conklyn Mini-Symposium, Department of Microbiology, Immunology and Molecular Genetics at UT Health, March 27, 2025.
6. Society for Mathematical Biology (SMB), Konkuk University, Seoul, Republic of Korea, June 30 - July 5, 2024.
7. Center-State Science Conclave, Vigyan Bhavan, Science City, Ahmedabad, inaugurated by Hon'ble Prime Minister of India, 10th -11th September 2022.
8. National Workshop on Technology Innovation in Cyber-Physical Systems (TIPS), National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS), DST, GoI, held at IIT Madras, May 6-7, 2022.
9. Rare Disease in Public Health: An Indian Context. DST-CPR, Indian Institute of Science (IISc), Bangalore Association with Ashoka University, February 19-20, 2022.
10. International Conference on Mathematical Computer Engineering-2020 organised by the Mathematics Division, School of Advanced Sciences, Vellore Institute of Technology, Chennai.
11. International Conference on Mathematical Techniques and Application-2020 sponsored by DST-SERB, SRM Institute of Science and Technology, Kattankulathur, Chennai.
12. Dr. Paarivendhar Research Colloquium – 2019 organised by Directorate of Research, SRM Institute of Science and Technology, Kattankulathur, Chennai.
13. Workshop on Applications of Mathematics in Engineering - 2019 conducted by SRM Institute of Science and Technology, Kattankulathur, Chennai.
14. Short Term Training Programme on Fuzzy sets and Applications - 2018 organised by Department of Mathematics, SRM Institute of Science and Technology, Kattankulathur, Chennai.
15. Franco-German Summer School on Inverse Problems and Imaging, September 18-22, 2017, Bremen, Germany.
16. Dynamical Developments: a conference in Complex Dynamics and Teichmüller theory, August 17-21, 2015, Bremen, Germany.

DELIVERED TALKs:

1. Vaccine Development Center of San Antonio Conference, 2025, United States.
2. Texas Biomed Research Symposium, 2025, San Antonio, United States.
3. International Conference on Mathematical Biology, Quantitative Modeling and Computation in Life Sciences and Medicine, June 12-14, 2025, Pokhara, Nepal.
4. 2025 Texas Biomed Research Symposium, Texas Biomedical Research Institute, April 4, 2025.
5. 2025 Conklyn Mini-Symposium, Department of Microbiology, Immunology and Molecular Genetics at UT Health, March 27, 2025.
6. Society for Mathematical Biology (SMB), Konkuk University, Seoul, Republic of Korea, June 30 - July 5, 2024.
7. International Conference & Symposium on AI Advancements in Hyperbolic and Parabolic PDEs (ICS-AIPDEs23), Department of Mathematics, University of Calicut, Kerala, India, and Department of Mathematical Science, UNIST, South Korea, 2023.
8. International Conference on Mathematical Techniques and Application-2020 sponsored by DST-SERB, SRM Institute of Science and Technology, Kattankulathur, Chennai, 2020.
9. International Conference on Mathematical Computer Engineering-2020 organised by the Mathematics Division, School of Advanced Sciences, Vellore Institute of Technology, Chennai, 2020.

EVENT ORGANISED:

1. Mini-symposium, Society for Mathematical Biology (SMB), Konkuk University, Seoul, Republic of Korea, June 30 - July 5, 2024.
2. Center-State Science Conclave, Vigyan Bhavan, Science City, Ahmedabad, inaugurated by Hon'ble Prime Minister of India, 10th -11th September 2022.
3. National Workshop on Technology Innovation in Cyber-Physical Systems (TIPS), National Mission on Interdisciplinary Cyber-Physical Systems (NM-ICPS), DST, GoI, held at IIT Madras, May 6-7, 2022.
4. Rare Disease in Public Health: An Indian Context. DST-CPR, Indian Institute of Science (IISc), Bangalore Association with Ashoka University, February 19-20, 2022.

LIST OF PUBLICATIONS:

1. **Saikat Batabyal,** Kevin Urdahl, Vitaly V Ganusov**:** Potential limitations of community-wide strategy to treat Mycobacterium tuberculosis infection. **(Communicated soon…)**
2. *Chakraborty, D, ***Saikat Batabyal,** Ganusov, V.**:** A brief overview of mathematical modeling of the within-host dynamics of Mycobacterium tuberculosis. **Frontiers in Applied Mathematics and Statistics**, 10, (2024). DOI: https://doi.org/10.3389/fams.2024.1355373. **(*Authors have equal contributions.)**
3. **Saikat Batabyal**: COVID-19: Perturbation dynamics resulting chaos to stable with seasonality transmission. **Chaos, Solitons & Fractals, (**SCI, Scopus, ISSN: 0960-0779, **IF: 9.922),** 110772, (2021).

4. **Saikat Batabyal**, Batabyal, A.: Mathematical computations on epidemiology: A case study of the novel coronavirus (SARS-CoV-2). **Theory in Biosciences, (**SCI, SCIE, Scopus, Online ISSN: 1611-7530, **IF: 1.315),** 140(2), 123-138, (2021).
5. **Saikat Batabyal**, Batabyal, A.: Public Healthcare System Capacity During COVID-19: A Computational Case Study of SARS-CoV-2. **Health Science Reports, (**PubMed, PubMed Central, Scopus, Online ISSN: 2398-8835), Article DOI: 10.1002/hsr2.305, (2021).
6. **Saikat Batabyal**, Jana, D., Upadhyay R. K.: Diffusion driven finite time blow-up and pattern formation in a mutualistic preys-sexually reproductive predator system: A comparative study. **Chaos, Solitons & Fractals, (**SCI, Scopus, Online ISSN: 0960-0779, **IF: 9.922)**, 147C, 110929, (2021).
7. **Saikat Batabyal**, Jana, D., Parshad, R. D., Basheer, A., Upadhyay R. K.: Pattern Formation in an Explosive Food Chain Model: The case of "Apparent" Mutualism. **Eur. Phys. J. Plus (**SCI, SCIE, Scopus, Online ISSN: 2190-5444, **IF: 3.758)**, 136**,** 448 (2021).
8. **Saikat Batabyal**, Jana, D., Lyu, J., Parshad, R.D.: Explosive predator and mutualistic preys: A comparative study. **Physica A (**SCI, Scopus, Online ISSN: 0378-4371**, IF: 3.778)**, (2019).
9. Jana, D., **Saikat Batabyal**, Lakshmanan, M.: Self-diffusion-driven pattern formation in prey-predator system with complex habitat under fear effect. **Eur. Phys. J. Plus (**SCI, SCIE, Scopus, Online ISSN: 2190-5444, **IF: 3.758)***,* 135, (2020).
10. Banerjee, D., **Saikat Batabyal**: Hyper relative order (p, q) of entire functions. **Annals of West University of Timisoara - Mathematics and Computer Science (**Online ISSN: 1841-3307), 2, 65- 84, (2017).
11. Dey, D., Kumar, A., Pandey, P., Batabyal, A., **Saikat Batabyal**, Lingayat, D.: Role of STI in Sustainable Development: Towards food and nutrition security. **Policy Brief, IISc, Bangalore**, (2022).